

Vylix Demo Material

Chapter 1: Limits and Continuity

The limit of a function $f(x)$ as x approaches a value c is the value that $f(x)$ approaches as x gets closer to c .

Definition: $\lim_{x \rightarrow c} f(x) = L$ means for every $\epsilon > 0$, there exists $\delta > 0$ such that if $0 < |x - c| < \delta$, then $|f(x) - L| < \epsilon$.

Key Theorems:

1. Squeeze Theorem
2. Intermediate Value Theorem
3. Mean Value Theorem